

Regression Testing

Meenakshi D'Souza

International Institute of Information Technology Bangalore.

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Overview

- Regression testing: Overview
- Regression test selection techniques
- Automation and tools

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Regression Testing

- Post release, a software is *maintained* by developers.
  - Maintenance involves making modifications to code in use, adding new code, removing code etc.
  - These changes are done in response to *change requests* by users/customers.
- Regression testing is the process of validating modified software to detect whether new errors have been introduced into previously tested code and to provide confidence that modifications are correct.
  - It is a black-box testing technique.
  - Typically very expensive.

Typical regression testing process

Let P be a procedure or program. Let P' be a modified version of P and let T be a test suite for P. A typical regression test proceeds as follows:

1. Select $T' \subseteq T$, a set of test cases to execute on P'.
2. Test P' with T', establishing P's correctness with respect to T'.
3. If necessary, create T'', a set of new functional or structural test cases for P'.
4. Test P' with T'', establishing P's correctness with respect to T''.
5. Create T''', a new test suite and test execution profile for P', from T, T' and T''.

Regression testing process: Corresponding problems

Each of the steps in regression testing leads to an important problem to solve.

- Step 1 involves the regression test selection problem: the problem of selecting a subset T' of T with which to test P'.
- Step 3 addresses the coverage identification problem: the problem of identifying portions of P' or its specification that require additional testing.
- Steps 2 and 4 address the test suite execution problem: the problem of efficiently executing test suites and checking test results for correctness.
- Step 5 addresses the test suite maintenance problem: the problem of updating and storing test information.

Regression vs. specifications

- Regression test selection is applicable both in cases where the specifications have not changed, and where they have changed.
- In the latter case, it is necessary to identify the test cases in T that are obsolete for P' prior to performing test case selection.
  - Test case t is obsolete for program P' iff t specifies an input to P' that is invalid for P' , or tt specifies an invalid input-output relation for P'.
- Having identified these test cases and removed them from T, regression test selection can be performed on the remaining test cases.

Selecting regression tests

Selecting regression tests

- Requires knowledge about the system and how it affects by the existing functionalities.
- Tests are selected based on the area of frequent defects.
- Tests are selected to include the area, which has undergone code changes many a times.
- Tests are selected based on the criticality of the features.

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Minimization techniques

- Minimization test selection techniques for regression testing attempt to select minimal sets of test cases from T that yield coverage of modified or affected portions of P.
- The problem, in its generality, is undecidable.
- 0-1 integer programming based techniques are available for procedures that are basically a sequence of statements with single entry and single exit (no branching, looping etc.).

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Dataflow techniques

- Data-flow coverage based regression test selection techniques select test cases that exercise data interactions that have been affected by modifications.
- For example, the technique of Harrold and Soffa [1988] requires that every definition-use pair that is deleted from P, new in P', or modified for P' be tested.
  The technique selects every test case in T that, when executed on P, exercised, deleted or modified definition-use pairs, or executed a statement containing a modified predicate.

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Safe techniques

- Techniques that are *not safe* can fail to select a test case that would have revealed a fault in the modified program.
- In contrast, when an explicit set of *safety conditions* can be satisfied, safe regression test selection techniques guarantee that the selected subset, T', contains all test cases in the original test suite T that can reveal faults in P'.
- For example, the technique of Rothermel and Harrold [1997] uses CFG representations of P and P', and test execution profiles gathered on P, to select every test case in T that, when executed on P, exercised at least one statement that has been deleted from P, or that, when executed on P', will exercise at least one statement that is new or modified in P' (when P is executed, a statement that does not exist in P cannot be exercised).

Random techniques

- In this case, a randomly selected set of test cases for P are executed on P'.
- This is often useless and done only to show that some regression testing is done.

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Retest-All techniques

- The retest-all technique simply reuses all existing test cases.
- To test P', the technique effectively *selects* all test cases in T.

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Regression test selection

- Different regression test selection techniques create different trade-offs between the costs of selecting and executing test cases, and the need to achieve sufficient fault detection ability.
- It has been found through empirical studies that the choice of regression test selection algorithm significantly affects the cost-effectiveness of regression testing.

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Automation of regression testing

- Most of the testing efforts in regression testing can be automated.
- Typically the previously executed test cases or the selected test cases are again run on a new build.
- Most of the regression test tools are record and playback type.
- The test cases are recorded by navigating through the program under test and verified whether expected results are coming or not.

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Some regression testing tools

- Regression Tester: Available from `http://www.regressiontester.com/`.
- TimeShiftX: Available from `https://www.vornexinc.com/`.
- Tools from Micro Focus: Available from `https://www.microfocus.com`.

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Some references

- Tools: Links already provided, Last access October 2017.
- A nice survey paper on regression test selection:
  T. L. Graves, M. J. Harrold, J-M. Kim, A. Porter and G. Rothermel, An Empirical Study of Regression Test Selection Techniques, in ACM Transactions on Software Engineering and Methodology, 30(2), April 2001.

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